Highlights

Spatial adjustment of Fire Weather Index thresholds through cluster analysis with environmental variables for the Province of Córdoba, Argentina.

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- Spatial calibration of Fire Weather Index (FWI) thresholds using environmental clustering.
- Percentile-based approach improves agreement between FWI and observed fire activity.
- Regionalisation reduces systematic overestimation of fire danger in Córdoba, Argentina.
- Framework is suitable for operational implementation in heterogeneous environments.

Spatial adjustment of Fire Weather Index thresholds through cluster analysis with environmental variables for the Province of Córdoba, Argentina.*

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ABSTRACT

The Province of Córdoba (Argentina) exhibits a highly dynamic fire regime, characterised by strong seasonality, recurrent ignitions, and a high potential for extreme wildfire events. Although the Fire Weather Index (FWI) is currently used in operational fire danger assessment, evidence indicates a systematic overestimation of danger levels when standard thresholds are applied, potentially leading to "warning fatigue" among civil protection organisations.

The representativeness of the FWI is improved in this study through a spatially explicit calibration of its danger thresholds. To achieve this, the province was regionalised into seven environmentally homogeneous regions using k-means clustering based on climatic, topographic, vegetation, and anthropogenic variables. For each cluster, FWI thresholds were redefined using a local percentile-based approach. To ensure statistical robustness and independence, the dataset was partitioned: fire records from the 2023 and 2024 seasons were utilised as the training set for regional threshold calibration, while the 2025 fire season was reserved exclusively for validation.

Significant spatial variability in FWI distributions was demonstrated, confirming the necessity of region-specific thresholds. Robust evidence is provided by statistical evaluation for the superiority of accumulated Burned Area (BA) over Hotspot Counts as a validation metric, achieving a significantly lower RMSE (0.073 vs 0.113) and a higher R^2 value (0.460 vs 0.382) compared to filtered hotspots in the critical 60th–95th percentile range. Furthermore, the concentration of 70.7% of the total burned area (BA) above the 80th percentile validates this threshold as a robust indicator of extreme fire conditions in Córdoba. Overall, the proposed methodology provides a scientifically rigorous and operationally applicable framework for adapting the FWI to heterogeneous environments, enhancing its reliability as a decision-support tool for wildfire management.

Spatial adjustment of Fire Weather Index thresholds through cluster analysis with environmental variables for the Province of Córdoba, Argentina.

1. Introduction

The Fire Weather Index (FWI) is one of the most widely used systems for evaluating wildfire danger based solely on meteorological conditions. Developed in Canada during the 1970s, it has since evolved into a globally adopted framework, with numerous regional adaptations that reflect different ecological and climatic contexts. Originally conceptualised by the Canadian Forestry Service (CFS) as part of the Canadian Forest Fire Danger Rating System ([19]), the FWI integrates temperature, relative humidity, wind speed, and precipitation to estimate fuel moisture and potential fire behaviour.

The first documented applications of the FWI in Europe emerged in the late 1990s, when San-Miguel-Ayanz and colleagues incorporated the index into the European Forest Fire Information System (EFFIS), adapting it to Mediterranean fire regimes. In Oceania, [14] employed the FWI to explore interannual variability in Burned Area (BA) across Tasmania and its relationship to climate drivers. In Asia—particularly in China— [18] assessed the suitability of the FWI in the Daxinganling region, finding that it adequately represented local fire danger conditions. In South Africa, [21] discussed fire management in Cape fynbos ecosystems, highlighting the relevance of fire danger indices for planning prescribed burns. In South America, early research applications of the FWI were reported in Chile and Argentina; for example, [23] used the index to study fire risk in temperate Patagonian

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forests, demonstrating its usefulness for regional fire danger assessments.

More recently, [11] emphasised the need to reassess FWI thresholds—particularly the "extreme" category—across diverse climatic and ecological contexts. Their systematic review of studies published between 1990 and 2024 identified several key drivers of threshold variability, including geographic setting, prevailing climate, climate variability, and anthropogenic influence. For example, in boreal forests, values of FWI > 30 frequently denote extreme danger, whereas in tropical savannas, higher thresholds (> 45) are more appropriate due to the fire-adapted nature of local vegetation. The authors also noted performance limitations: the FWI tends to underestimate danger in regions with strong seasonal humidity (e.g., Southeast Asia) and to overestimate it in landscapes dominated by unaccounted-for wind patterns (e.g., Mediterranean regions). Furthermore, about 60% of the areas analysed showed an increasing frequency of extreme-FWI days in the last decade, particularly in the Arctic and Amazon basin. The review also demonstrated that, in human-modified landscapes (e.g., agricultural regions of Sub-Saharan Africa), the FWI benefits from adjustments incorporating ignition sources or landscape fragmentation.

In South America, research evaluating the performance of the FWI remains limited. In Argentina, early fire danger assessments relied on single-index meteorological indicators ([6]), and the country lacked a coordinated national policy for fire management. A series of severe wildfires in Patagonia during the 1980s—culminating in the tragic 1994 event near Puerto Madryn, Chubut, where twenty-four firefighters lost their lives—triggered the development of national fire management strategies. These efforts advanced understanding of fire danger and promoted the adoption of decision-support technologies.

Formal institutional actions began with "Proyecto 2000", a joint initiative between the British Columbia Forest Service and Argentina's National Fire Management Plan (established in 1996). The project aimed to operationalise a fire danger rating system in pilot areas of Patagonia, including the Andean region of Chubut Province and Lanín National Park in Neuquén Province. Later studies by [3] analysed ignition probability in dead fine fuels within Patagonian cypress (Austrocedrus chilensis) and ñire (Nothofagus antarctica) forests, ultimately calibrating the Fine Fuel Moisture Code (FFMC) for the region. They identified the following FFMC categories: low (≤ 85; MC ≈ 38–39%), high (85–89; MC ≈ 6–11%), and extreme (> 89; MC < 6%).

Today, the National Meteorological Service (SMN) of Argentina provides daily FWI estimates based on station observations, supporting provincial and municipal agencies. In the province of Córdoba, the FWI has been used operationally since 2011. In 2021, the Córdoba Hydrometeorological Observatory (OHMC), in collaboration with the Provincial Fire Management Plan and at the request of the Secretariat for Climate Risk Management and Civil Protection, initiated the development of a numerical-model-based FWI system using WRF simulations ([17, 15, 13]). This system has been operational since mid-2022 and produces daily FWI maps for central Argentina, particularly Córdoba Province, accessible online (https://mapascordoba.gob.ar/viewer/mapa/442). The implementation also provides 24-hour forecasts of FWI components, including fuel moisture codes (FFMC, DMC, DC) and behaviour indices (ISI and BUI), enhancing regional fire risk preparedness.

In the province of Córdoba, the OHMC currently generates operational fire danger maps using the FWI calculated from high-resolution WRF model simulations. However, operational experience and preliminary data indicate a systematic tendency of this system to overestimate fire danger. This phenomenon of overestimation via standard thresholds has been documented in analogous regions such as southern Greece [22] and the United Kingdom [10], where it resulted in a proliferation of "false alarms" and subsequent "warning fatigue" among civil protection organisations. These inaccuracies often stem from the structural limitation of the FWI's structural limitation as a "weather-only" index [19], which assumes a standardised fuel load and flat terrain, thereby ignoring the decisive influence of topography and local biomass heterogeneity on fire spread. Following the recommendations on the need to reassess thresholds across diverse ecological zones [11], this study proposes a local calibration of the FWI through cluster analysis [12]. By integrating climatic, topographic, vegetation, and anthropogenic factors, and replacing fixed numerical thresholds with an approach based on local percentiles, this work aims to establish danger categories that more faithfully reflect the actual risk of ignition and spread. Finally, these new categories are validated using satellite-derived hotspots as 'ground truth' data to ensure their operational reliability across Córdoba's diverse territory [4].

In this context, the main objective of this study is to improve the representativeness of the FWI in the Province of Córdoba through a spatially explicit calibration of its danger thresholds. By integrating environmental heterogeneity via cluster analysis and redefining thresholds based on local percentiles, this work seeks to reduce systematic biases in fire danger estimation and enhance its operational applicability. To achieve this, the paper is structured as follows: Section 2 describes the environmental and fire-regime characteristics of the study area; Section 3 presents the datasets and modelling framework; Section 4 details the methodology, including clustering and percentile-based threshold estimation; Section 5 discusses the main results; and Section 6 summarises the conclusions and implications for fire danger assessment.

2. Characteristics of the Province of Córdoba in Relation to Wildfires

The province of Córdoba presents a combination of biophysical and socio-territorial factors that shape a fire environment characterised by high recurrence, marked seasonality, and the potential for extreme wildfire behaviour.

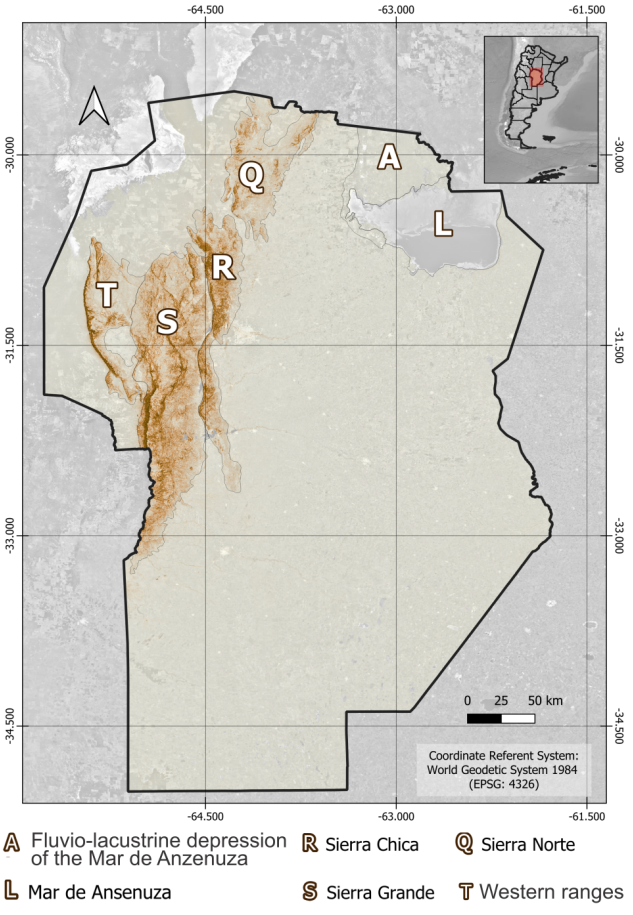


Figure 1: Geographical and topographic characterisation of the Province of Córdoba, Argentina. The map illustrates the complex relief and the main mountainous systems that drive local wildfire dynamics: (R) Sierra Chica range, (S) Sierra Grande range, (Q) Sierra Norte range, and (T) Western ranges. Also identified are (L) the Mar de Ansenuza and its (A) fluvio-lacustrine depression.

A defining element of Córdoba's fire regime is the predominance of highly flammable vegetation types, including shrublands, grasslands, and mountain forests distributed across the Sierras Grandes and Sierras Chicas as shown in Figure 1. These ecosystems accumulate significant fine and coarse fuels during winter and early spring, when precipitation is limited and atmospheric humidity decreases.

Consistent with this seasonal drying, the *Informe Anual de Áreas Afectadas por Incendios Forestales 2025* [9] describes a marked increase in wildfire activity during the late winter–spring period, a pattern also evident in the peak number of fires observed between August and October in Figure 2. Additionally, topographic complexity—particularly the steep slopes characteristic of the mountainous systems—further influences fire behaviour by enhancing wind exposure, promoting rapid upslope spread, and complicating suppression operations. The large burned-area peaks observed in certain years, especially in late winter and early spring, illustrate how these environmental and anthropogenic drivers combine to shape Córdoba's highly variable but recurrent fire regime.

Climatic variability further modulates the fire environment. Interannual fluctuations in precipitation and temperature—often linked to large-scale modes such as ENSO—affect fuel moisture and the frequency of high fire-danger days. These characteristics are evident in the monthly evolution of wildfire occurrence and BA between 2022 and 2025 [9](Figure 2), which shows a consistent concentration of events during the austral winter–spring period. Operational records for 2022–2025 indicate that some years exceed 177 documented ignitions per month during the peak season, underscoring the strong coupling between meteorological drivers and ignition likelihood.

In particular, both the number of fires and the affected area increase markedly from June onwards, reaching peak activity between August and October. The year 2024 stands out for its extreme behaviour, with a pronounced peak in BA in September, while 2025 exhibits a high number of fire detections but comparatively lower BA, suggesting differences in fire size or duration. In contrast, the summer months show relatively low fire activity across all years.

This seasonal pattern reflects the combined influence of fuel availability, prolonged dry conditions, and meteorological factors that favour fire ignition and spread, reinforcing the role of Córdoba as a region with recurrent and highly variable wildfire dynamics.

Human influence constitutes an additional structural driver. According to Argañaraz and Bellis [1], most wildfire ignitions in Córdoba originate from anthropogenic sources, including agricultural burning, negligent behaviour, intentional fires, and the expansion of the wildland–urban interface. These ignition pressures interact with the inherent flammability of the landscape and contribute to the spatial clustering of fire activity observed throughout valleys, foothills, and peri-urban corridors.

3. Materials

3.1. Fire Weather Index (FWI)

The FWI System was first issued in 1970 after several years' work by several fire researchers in the Canadian Forestry Service. The FWI is based on the moisture content of three classes of forest fuel plus the effect of wind on fire behaviour. The system consists of six components: three primary subindexes representing fuel moisture content, two intermediate subindexes representing rate of spread and fuel consumption, and a final index representing fire intensity as energy output rate per unit length of fire front. Its components are determined every day from noon weather readings: temperature, relative humidity, wind speed, and rain (if any) [19].

Figure 3 presents a summarised flowchart of the Fire Weather Index (FWI) calculation procedure. Meteorological conditions are first obtained from the WRF model, including 2-m air temperature (T2), 2-m relative humidity (RH2), 10-m wind speed (10m-Wind), and accumulated precipitation

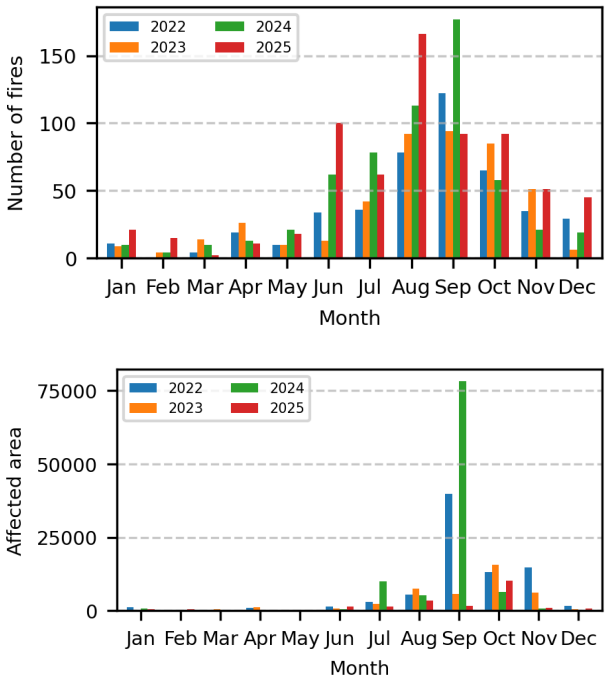


Figure 2: Monthly evolution of wildfire occurrence and BA in the Province of Córdoba for the period 2022–2025. The figure shows the seasonal concentration of fire activity, with a marked increase during the austral winter–spring period and interannual variability in both the number of events and the affected area.

over the previous 24 h (PRECIP). These variables are used to compute the FWI for the subsequent 24 h forecast period.

The first step consists of updating the Fine Fuel Moisture Code (FFMC), Duff Moisture Code (DMC), and Drought Code (DC). At this stage, copies of the previous-day FFMC, DMC, and DC are created and denoted as F_0, P_0, and D_0, respectively. This allows the calculation to explicitly account for both the antecedent (previous day) and current-day fuel moisture conditions. The codes for the current day are then updated by evaluating whether accumulated precipitation during the last $24h$ exceeds predefined threshold values. Depending on whether these thresholds are met, different sets of equations are applied in the subsequent calculations.

For the DMC and DC, a day-length adjustment factor is introduced. This factor is a monthly numerical value that represents the variation in daylight duration throughout the year and accounts for seasonal effects on fuel drying. Once the fuel moisture codes have been updated, the Initial Spread Index (ISI) is computed as a function of the FFMC and wind speed, representing the potential rate of fire spread following ignition. In parallel, the Buildup Index (BUI), which integrates information from both the DMC and DC, is calculated to describe the amount of fuel available for combustion.

Finally, FWI is derived from the ISI and BUI, providing an integrated measure of potential fire intensity expressed as the energy output per unit length of the fire front. For a more detailed description of the calculations, the reader is referred to the official documentation provided in [19] and [20].

Table 1
Physical parameterisation schemes used in the WRF configuration

Model component	Parameterisation scheme	Option
Microphysics	Thompson microphysics	8
Planetary boundary layer	Mellor-Yamada-Janjic scheme	2
Shortwave radiation	Dudhia scheme	2
Longwave radiation	RRTM scheme	1
Land surface model	Unified Noah land surface model	2
Surface layer	Eta similarity scheme	2

3.2. The Weather Research and Forecasting Model (WRF)

The WRF model, version 4.2, is employed in this analysis [16, 8]. Initial and boundary conditions are obtained from the 06:00 UTC run of the Global Forecast System (GFS), which has a horizontal resolution of 0.25°. The WRF model domain consists of 270 × 270 grid points, with a horizontal resolution of 4 km and 35 vertical levels. Table 1 summarises the physical parameterisation schemes used.

The indices are computed daily at 13:00 local time (16:00 UTC) using custom Python scripts. Each script execution produces the forecast Fire Weather Index (FWI) for the following day. The resulting indices are then visualised and exported in both PNG and GeoTIFF formats for further analysis.

3.3. Burned Areas

The input data for the analysis model are derived from two main sources: Burned Area polygons (BA) and data on thermal anomalies, known as hotspots, based on satellite information.

The BA maps correspond to the areas actually affected by forest and rural fires. The data used comes from the official mapping of the Government of the Province of Córdoba, published by the Risk Management Directorate (Ministry of Security – Secretariat for Climate Risk Management, Disasters, and Civil Protection) and available on the Spatial Data Infrastructure (SDI) geoportal of the Province of Córdoba – IDECOR (`https://mapascordoba.gob.ar/viewer/mapa/442`). Access to the data was obtained through a connection to WFS geoservices provided by the provincial SDI. For this analysis, the time series corresponding to the 2023–2025 period was used, which includes forest and rural fire events throughout the province.

Generation of maps was developed by IDECOR's technical team, based primarily on the BA Mapping Tools (BAMT v1.7) [2] implemented in Google Earth Engine. The procedure relies on supervised classification using the Random Forest algorithm, trained with spectral variables derived from post-fire imagery and multitemporal difference images (e.g., NBR and dNBR). The classifier assigns each pixel a probability of belonging to the "burned" class (0–100%).

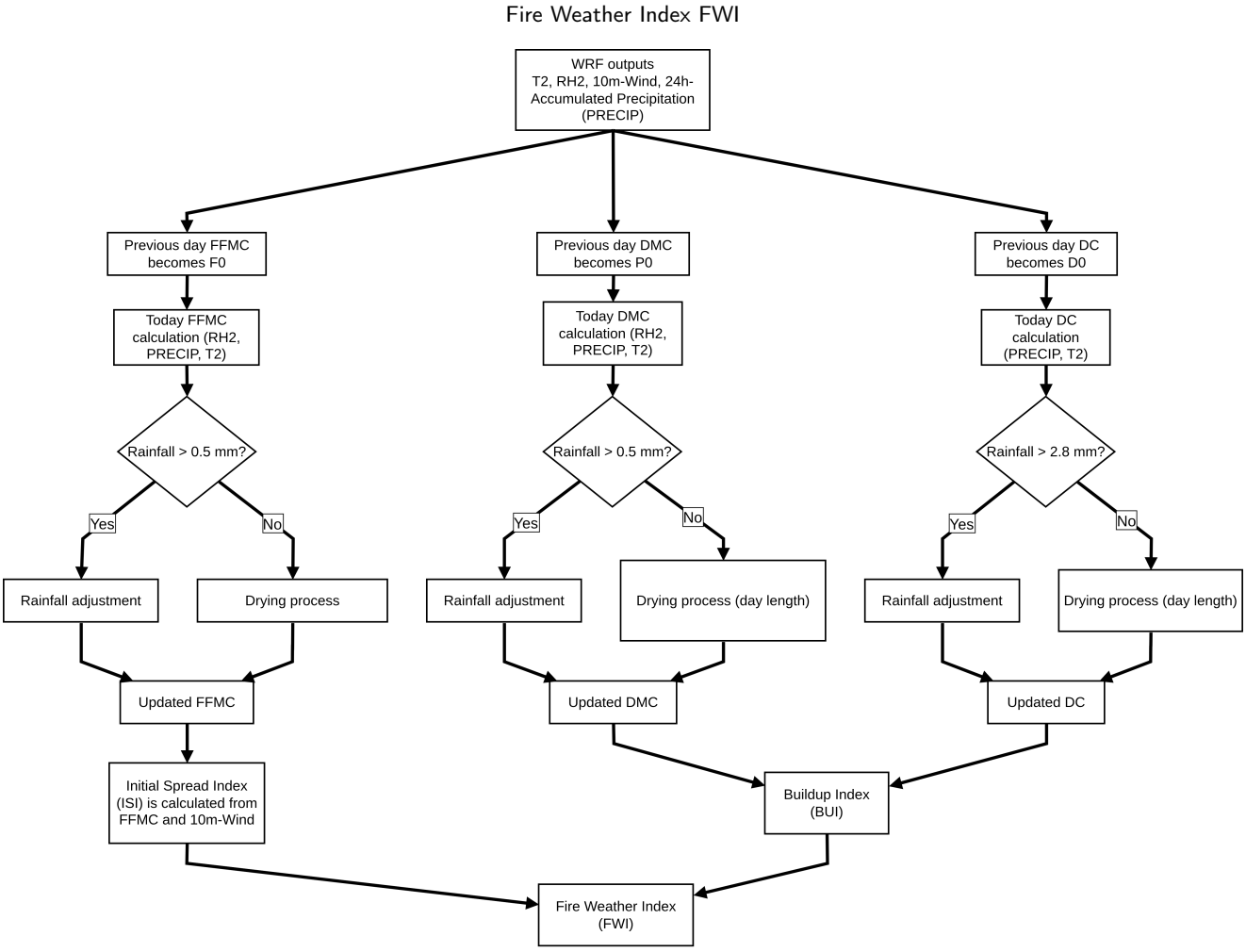


Figure 3: Simplified flowchart of the Fire Weather Index (FWI) system forced by WRF outputs. Daily meteorological variables simulated by the WRF model are used to compute the fuel moisture codes (FFMC, DMC, and DC), which are then combined to derive the Initial Spread Index (ISI), the Buildup Index (BUI), and the final FWI.

The final result consists of a binary map of BA with an associated detection date and an estimate of the affected area. The input spectral data came from the Sentinel-2 satellite constellation (spatial resolution of 10/20 m and revisit frequency of 5 days) and Landsat 8 and Landsat 9 (spatial resolution of 30 m and effective revisit of 8 days when combined). The training data (samples) are generated from official fire records, georeferenced and validated by the Secretariat for Risk, Climate Change, and Civil Protection (SRCCyPC), through the Communications Centre Information System (SICC), and by the Directorates of Resource Management (DGR) and Integrated Fire Management (DirGIMF).

One option to obtain daily records is to use the Fire Information for Resource Management System (FIRMS) database. Thermal anomaly products were used from: VIIRS (C2), aboard the Suomi-NPP, NOAA 20 (J1), and NOAA 21 (J2) satellites, and from MODIS Collection 6.1, corresponding to the period from January 1, 2023, to December 31, 2025. The data were used primarily in their standard version (not near-real time). These datasets are processed by the University of Maryland and distributed through FIRMS with an approximate three-month lag, replacing the near-real-time (NRT) files. The data are provided in the WGS84 geographic projection, a system that was retained for subsequent analysis.

4. Methodology

4.1. Preparation of the temporal sequence

For this study, it was necessary to establish a temporal link between the occurrence of BA and the daily meteorological conditions used to calculate the FWI, as well as to estimate more accurately the area actually affected on each day. The dataset was partitioned into two distinct periods: fire records from both the 2023 and 2024 seasons were used as the training set for regional threshold calibration, while the 2025 fire season was reserved exclusively for independent validation.

Thus, for each BA polygon, the probable start date of the event was reported, as obtained from the aforementioned official records. If this information was not available, the date was assigned based on fire hotspot data (FIRMS) or, as a last resort, on the date of the first satellite image in

which the burn scar was detected. In this way, all polygons were assigned a single date. However, for large-scale fires, information was not available for the subsequent days during which the fire remained active until its final extinction.

4.2. Filtering and integration of BA and Hotspot data

The study used a database representative of the entire study area in Córdoba Province, to which data filtering criteria were applied in order to remove false detections associated with high temperatures unrelated to fires, such as industrial sources, sun-exposed surfaces, and landfill burning, as well as possible omissions caused by cloud cover, dense smoke, low-intensity fires, or events below the sensor's detection threshold. Additional sources of uncertainty included potential geolocation errors and limitations inherent to the sensors' spatial and temporal resolution, which may result in multiple detections of the same incident or, conversely, in the failure to record small-scale hotspots. False positives were filtered out by spatially associating heat source detections with the BA polygons from the aforementioned official mapping. Based on this cross-referencing, heat sources coinciding with burn scars were identified, taking into account whether the dates of the thermal anomalies fell within a time window close to the start date assigned to the corresponding polygon.

As heat source detection data derived from MODIS and VIIRS have spatial resolutions of approximately $375m$ and $1km$, respectively, while FWI has a spatial resolution of $4km$; to create a single daily dataset, a hexagonal grid with an intermediate spatial resolution of $2km$ was adopted. This strategy addressed the well-known problem of spatial resolution mismatch, which can introduce bias when combining datasets with different observation scales. By defining units equivalent to one quarter of the FWI pixel area, it was possible to characterise sub-pixel fire variability without introducing artificial detail into the explanatory climatic variable [4].

The hexagonal tessellation was selected for its favourable geometric properties in spatial analysis, as it minimises edge effects and directional biases, ensuring uniform neighbourhood relationships [24]. This approach provided a more stable and statistically consistent representation of the spatial variability of heat sources within the meteorological context defined by the FWI.

Hotspots were assigned to the hexagonal grid using a nearest-neighbour interpolation method that preserves the original values. In this way, each hexagon within the BA was assigned a daily date of fire activity, enabling the temporal evolution of the fire and the affected area on each day of the event to be estimated [7].

To illustrate this methodology, Figure 4 shows a wildfire that occurred near the town of Villa Carlos Paz, which began on 9 October 2023 and affected a total area of approximately 3,714 ha. The figure combines the BA polygon with FIRMS hotspot detections, differentiated by date, allowing the persistence and spread of the event between 9 and 11 October 2023 to be identified. It also shows the hexagonal grid used as the spatial unit for temporal assignment and the resulting estimation of daily BA.

4.3. Clustering

Given the marked environmental heterogeneity of Córdoba, ranging from humid plains to semi-arid mountain ranges, the territory was subdivided into homogeneous regions to improve the local representativeness of the FWI. For this process, an initial set of 60 numerical variables was selected, representing the main environmental factors that influence wildfire occurrence and spread across the province. These factors encompassed hydroclimatic records (1981–2024), topography, soil properties, hydrology, land cover (2000–2024), and anthropogenic infrastructure.

To reduce redundancy and mitigate the influence of multicollinearity in the clustering process, a linear correlation analysis was performed using Pearson's correlation coefficient. For pairs of variables exhibiting an absolute correlation greater than 0.90, a statistical filter was applied to retain only those most representative of long-term environmental conditions. As a result of this procedure, four specific variables were disregarded and excluded from the final set: EVI Index (median of the MODIS 2000–2024 time series), mean annual potential evapotranspiration (ERA5-Land 1981–2024), mean annual minimum temperature (ERA5-Land 1981–2024), and mean annual temperature (ERA5-Land 1981–2024).

Following this reduction, a set of 56 variables representative of the main environmental factors influencing wildfire occurrence and spread in Córdoba Province was selected and organised into the following categories (numbers in brackets indicate the number of variables associated with each environmental group):

- **Hydroclimatic variables (1981–2024) [10]:** Annual actual and potential evapotranspiration, water deficit, Palmer Drought Severity Index (PDSI), annual precipitation, annual wind speed, annual mean, maximum, and minimum temperatures, and annual solar radiation, derived from ERA5-Land reanalysis data [5].
- **Topographic variables [2]:** Elevation and slope.
- **Edaphic variables [8]:** Soil productivity and physical soil properties, including organic matter levels, phosphorus content, organic matter storage, pH levels, and other granulometric variables.
- **Hydrological variables [3]:** Distance to watercourses, presence of permanent surface water bodies, and historical recurrence of surface water.
- **Land cover, vegetation, and anthropogenic variables [24]:** EVI and NDVI vegetation indices (2000–2024). In addition, the distribution of 22 land cover and land use categories was included (Land Cover 2022/2023, IDECOR, 2023).

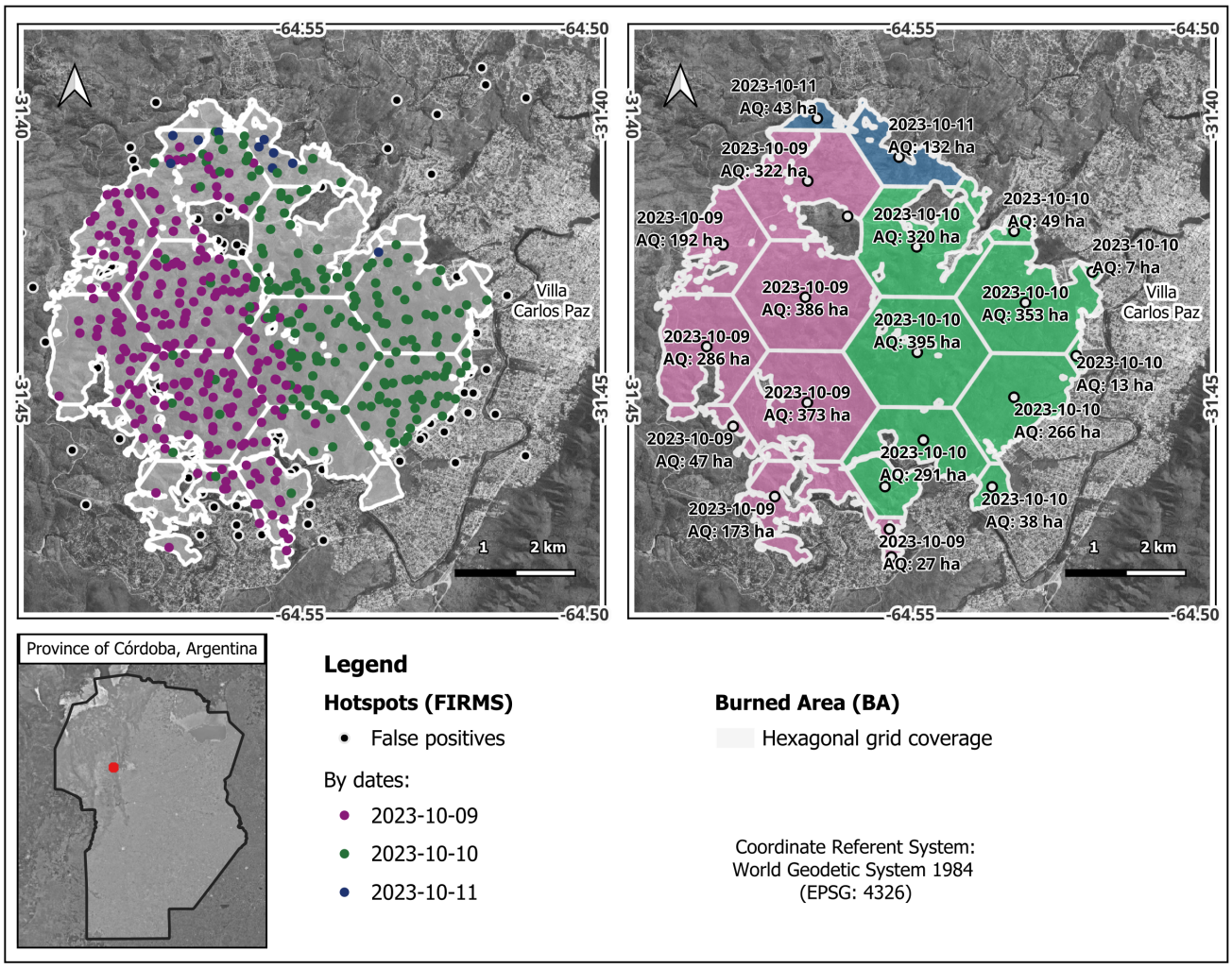


Figure 4: Temporal characterisation of the forest fire that occurred near the town of Villa Carlos Paz in October 2023, based on the integration of hotspot detections and BA. The left-hand side shows the original hotspots for three different dates, whereas the right-hand side shows the final image following the rasterisation process.

- **Fire-related variables [5]:** Fire occurrence during the 2022 to 2025 period. In addition, a normalised fire hazard index was included.
- **Infrastructure variables [4]:** Distance to urbanised areas, power transmission networks, and the paved and unpaved road access network.

Given that the variables used are expressed in different units and scales, such as millimetres, degrees, and metres per second, they were standardised before analysis. Each variable was transformed by subtracting its mean and dividing by its standard deviation. This procedure ensured that all variables contributed comparably to the distance calculations performed during the clustering process and prevented those with a larger numerical range from dominating the results. Subsequently, records containing missing values in any of the selected variables were removed, so that only grid cells with complete information were retained.

The spatial grouping of the units, that is, the identification of spatial patterns, was performed using the unsupervised *k-means* clustering algorithm [12]. This method partitions the observations into k groups by minimising within-cluster variability based on Euclidean distances in the space of standardised variables. Clustering quality was assessed using the *Silhouette* coefficient, which quantifies the degree to which each observation resembles its own cluster relative to neighbouring clusters. This index ranges from -1 to 1, with higher values indicating greater within-cluster cohesion and better separation between groups.

4.4. Redefinition of thresholds (percentile-based estimation)

Following the methodology proposed by [22] for the regional adaptation of the FWI in Greece, this analysis implemented an approach based on Percentile Indices (PI) applied to each of the seven regions defined through clustering. This approach made it possible to establish fire danger thresholds that are specific to, and representative of, the

relative extreme meteorological conditions of each *cluster*, thereby overcoming the limitations of a single classification for the entire province.

The analysis period comprised the months of the fire season, from June to November, for the years 2023 and 2024. For this period, daily FWI rasters generated by the WRF meteorological model at a spatial resolution of 4 km were used.

Prior to percentile estimation, a spatial filter was applied using a vegetation mask derived from the land cover map. This mask was constructed through the following steps:

1. **Exclusion of non-combustible areas:** Urban areas, water bodies, and salt flats were removed.
2. **Size filtering:** All fragments of the excluded land covers with an area smaller than 1600 ha, equivalent to the area of one FWI pixel, were discarded in order to avoid a "speckled" effect and excessive clipping.
3. **Rasterisation:** The mask was rasterised at a resolution of 3.5 km, slightly finer than that of the FWI (4 km), thereby optimising spatial overlap and minimising the loss of vegetated pixels along the edges of the excluded areas.

This mask made it possible to retain only those pixels corresponding to areas with potentially combustible vegetation, while excluding those located in areas where the calculation of fire danger lacks operational relevance.

4.4.1. Cluster-Based Processing and Percentile Calculation

Data processing was carried out using an automated script which, for each of the clusters, performed the following operations:

1. **Accumulation of FWI values:** For each cluster i ($i = 1, \ldots, n$), all daily FWI values corresponding to the June–November period of 2023 and 2024 were extracted and accumulated. Pixel selection was restricted to those meeting two conditions: (a) being included in the vegetation mask, and (b) spatially belonging to the polygon of cluster i.
2. **Percentile calculation:** Based on the accumulated values, FWI percentiles from 0 to 100 were calculated at intervals of 5 units (0, 5, 10, …, 95, 100). This comprehensive analysis made it possible to characterise the full distribution of the index in each of the n environmentally homogeneous regions.
3. **Validation using fire data:** To determine which percentiles were most appropriate as danger thresholds, two datasets were systematically compared against the previously calculated percentiles for each cluster:
   - FWI values associated with each hotspot at the time of detection;
   - FWI values corresponding to BA on the day of ignition.

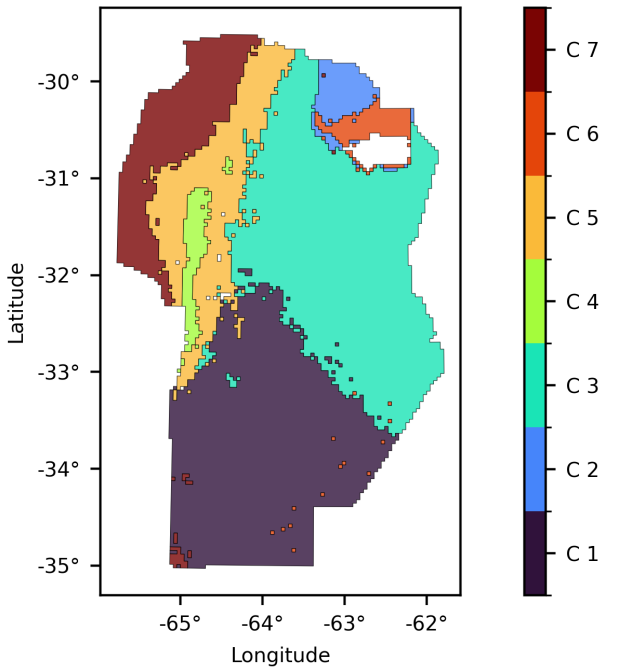


Figure 5: Environmental regionalisation of Córdoba Province using *k-means* clustering. The seven identified groups capture spatial variability in climate, topography, vegetation, and anthropogenic factors across the province.

Following the approach proposed by [22], the cumulative percentage of hotspots and BA occurring when FWI exceeded each percentile (P_k) was analysed. This relationship made it possible to identify the percentiles showing the strongest association with actual fire occurrence.

5. Results and discussions

5.1. Clustering

The statistical quality of the regionalisation is evaluated for configurations of k = 5, 6, and 7. The mean Silhouette coefficients obtained are 0.185, 0.196, and 0.205, respectively, identifying k=7 as the local mathematical optimum. This value reflects the biophysical reality of the Province of Córdoba, where environmental variables follow continuous gradients, particularly the humidity and altitudinal transitions between the eastern plains and the mountain ranges, rather than forming discrete, isolated units [5].

According to this criterion, the study area is partitioned into seven clusters, as seen in Figure 5. This result indicates a low to moderate separation between groups, suggesting the presence of continuous environmental gradients rather than clearly delimited discrete units. Nevertheless, despite its limited quality from a strictly statistical standpoint, the resulting clustering is suitable for exploratory purposes and for identifying dominant environmental patterns at the regional scale.

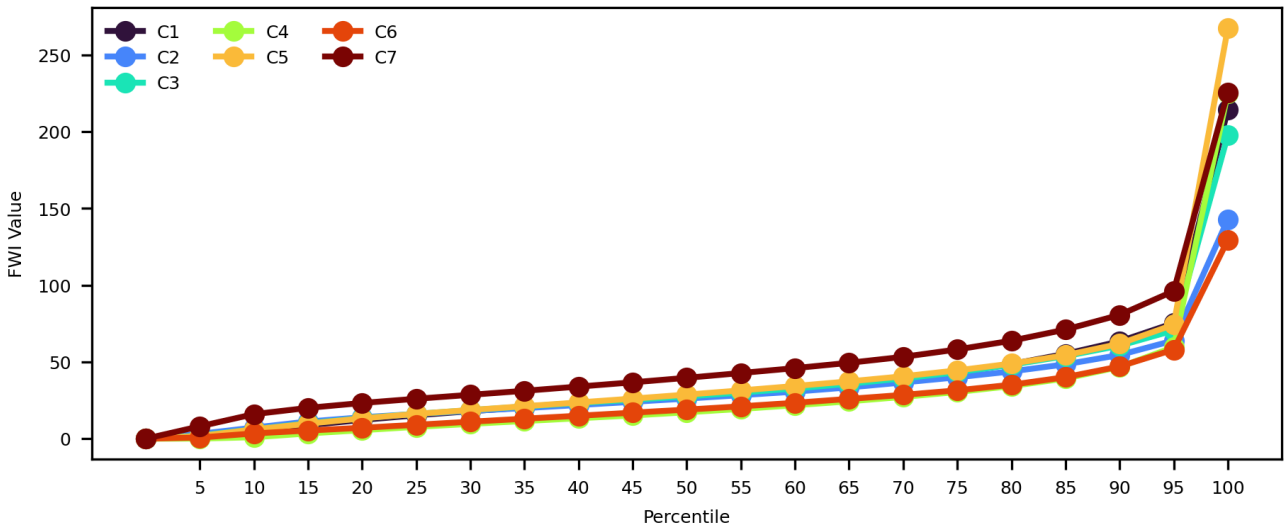


Figure 6: Distribution of FWI percentiles by cluster.

5.2. Thresholds and Percentiles

Figure 6 presents the percentile distribution curves of the FWI for each of the seven clusters identified in Córdoba. These curves reveal differentiated patterns between the environmentally homogeneous regions.

The distribution of Cluster 7 shows the steepest slope, reaching FWI values above 80 at the 90th percentile, indicating typically more extreme weather conditions in this region. In contrast, Cluster 4 presents the flattest curve, with a 90th percentile of approximately 47, reflecting less severe FWI behaviour. Clusters 1, 3, and 5 exhibit intermediate but similar behaviours, with 90th percentiles around 60–64, while Cluster 2 shows slightly lower values in the high percentiles. This comprehensive visualisation demonstrates that the clusters differ not only in specific values but in the overall shape of the FWI probability distribution, reinforcing the relevance of a reclassification adapted to each regional context.

To determine which percentiles are most suitable as boundaries between hazard categories, the relationship between FWI percentiles and two response variables is evaluated: the number of hotspots and the cumulative BA. The distinction between the two is not trivial: hotspots (derived from satellite sensors such as *VIIRS* or *MODIS*) indicate the occurrence of ignitions, but not necessarily the magnitude of the fire; by contrast, the BA represents actual fire behaviour and its propagation capacity, constituting a more direct indicator of hazard.

Figure 7 illustrates the statistical relationship between regional FWI percentiles and observed fire activity in Córdoba, showing the distribution of filtered hotspots in the upper panel and accumulated burned area (BA) in the lower panel across the full percentile range (0–100th). Both panels show a focused regression analysis of the critical region between the 60th and 95th percentiles, where fire activity begins to escalate significantly.

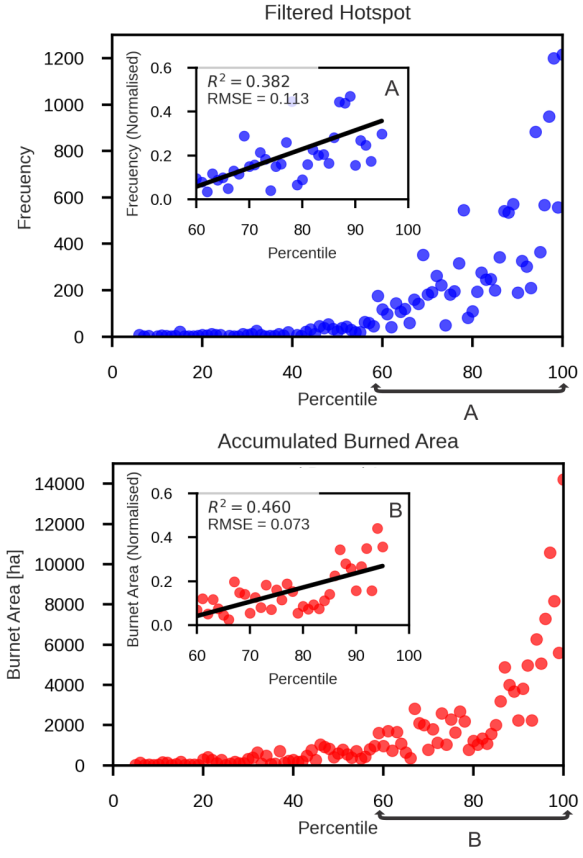


Figure 7: Relationship between FWI percentiles and fire activity indicators.

The low-to-mid percentiles (0–60th), are not relevant for BA analysis. This reflects "ignition noise" thermal anomalies primarily driven by anthropogenic activities (such as

small-scale agricultural burning or landfill fires) which occur independently of extreme fire weather.

Conversely, the BA achieves a better statistical fit in the 60th–95th range, with a coefficient of determination (R^2 = 0.460) and a lower Root Mean Square Error (RMSE = 0.073) compared to filtered hotspots (R^2 = 0.382; RMSE = 0.113). These results confirm that BA acts as a physical filter, effectively isolating the fire-weather signal from anthropogenic noise and ensuring that the new regional thresholds are calibrated to identify events with high propagation potential, which pose the greatest challenge to suppression resources. The variance associated with R^2 for BA can be explained from the inherent standpoint of the nature of the FWI, given that it depends solely on meteorological factors, as it quantifies the potential fire intensity and spread rate assuming a standardised fuel load and flat terrain. Consequently, the residual variance can be attributed to three primary systematic factors that characterise the fire regime in Córdoba: stochastic human ignitions, topographic and fuel complexity, and the active work of firefighters, since a fire that ignites under extreme FWI conditions may result in a minimal BA if it is contained quickly.

Based on the results obtained, BA was selected as the primary reference variable for defining hazard thresholds, given its more direct relationship with fire impact. To establish these benchmarks, the FWI distribution for the entire province is integrated and compared against all observed burned areas. Figure 8 illustrates the distribution of cumulative BA as a function of these province-wide FWI percentile intervals. This figure reveals a strongly non-linear relationship between fire impact and FWI percentiles. For percentiles below P20, the contribution to the total BA is negligible (less than 0.5%). Between P20 and P60, a gradual increase is observed, reaching approximately 9.6% of the cumulative BA. However, a marked transition occurs beyond P60, with a sharp increase between P60 and P80 (19.1%), followed by a dominant contribution above P80, which alone accounts for 70.7% of the total BA. This behaviour highlights the 80th percentile as a robust regional indicator of extreme fire conditions across the study area.

Based on this empirical relationship, the 20th, 40th, 60th, and 80th percentiles are selected as thresholds for the regionalised reclassification (Table 2). This selection ensures consistency between the statistical distribution of the FWI and the observed fire impact. Full values for all percentiles (0-100), computed at 5-percentile intervals, are available in Table 3 of Appendix A.1.

5.3. Redefinition of thresholds and validations

Data from the 2025 fire season are reserved as a validation set. While the regional FWI thresholds are calibrated using the June–November period of 2023 and 2024.

According to the environmental differences observed within the clusters, it is possible to define different levels of fire risk in the province of Córdoba, where "risk" is understood as the combination of the territory's biophysical

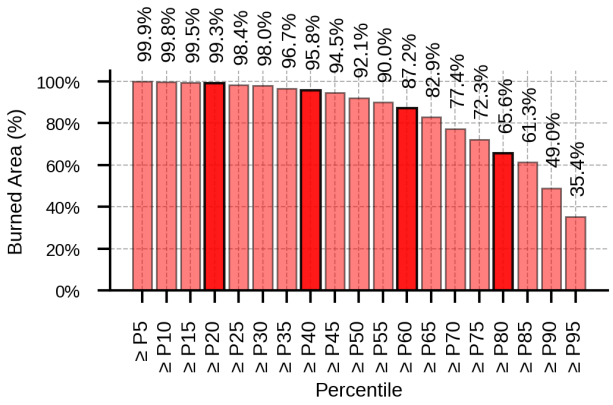


Figure 8: Cumulative BA as a function of FWI percentile intervals.

Table 2
FWI thresholds (20th, 40th, 60th, and 80th percentiles) for regionalised reclassification in Córdoba

Cluster	P20	P40	P60	P80
C 1	12.2	22.6	32.1	48.6
C 2	13.9	21.9	30.6	43.8
C 3	13.4	23.2	32.7	48.1
C 4	5.5	13.3	21.8	34.3
C 5	13.2	23.6	34.3	48.9
C 6	7.1	14.9	23.4	35.1
C 7	23.2	33.9	45.9	63.8

conditions, the probability of ignition, and the potential damage caused by fires.

Figure 9 presents, the comparisons following the grouping between the original FWI categories without using percentiles (orange bars) and the new categories obtained using percentiles. It should be noted that Clusters 2 and 6 were excluded from this validation because these regions are primarily characterised by controlled burning practices, for which official statistical fire records are currently not being generated in the Province of Córdoba.

In Cluster 1 (southern and south-western region), environments dominated by grasslands predominate, characterised by the continuity of fine fuels and frequent winds. These conditions favour fast-moving fires that can spread rapidly. In these sectors, the adjusted model increases the proportion of burnt area in the moderate categories and reduces its concentration in the high and extreme categories compared to the original model. Thus, the adjusted model corrects the tendency of the original model to overestimate the hazard in these grassland environments, where, although spread is rapid, it does not always result in high-intensity fires.

In Cluster 3 (the central-eastern region), agricultural practices predominate, with significant changes to the landscape throughout the seasons. Consequently, the observed hazard is relatively lower and better controlled, due to the limited availability of natural fuels and their spatial discontinuity. The fires that occur are usually localised and of low

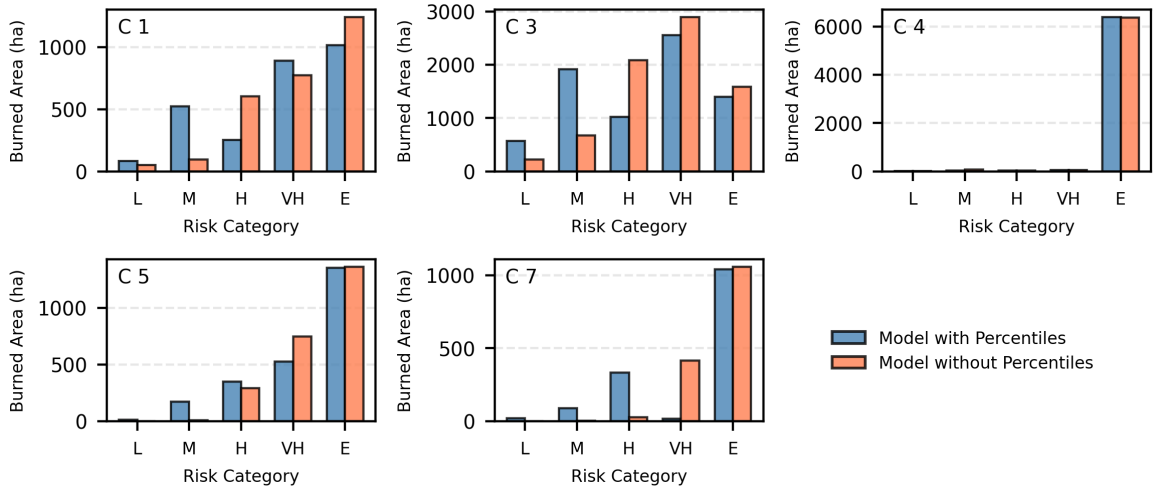


Figure 9: Validation of the FWI model. Comparison between classification with percentiles (proposed) and without percentiles (standard). Accumulated BA by risk category is shown for each cluster. Clusters 2 and 6 were excluded from this validation because these regions are primarily characterised by controlled burning practices, for which official statistical fire records are not currently generated by the Province of Córdoba.

intensity, generally associated with agricultural practices, where the risk is predominantly human-induced. As for Cluster 1, the adjusted model, compared with the original, tends to significantly increase the proportion in the low and moderate categories, reducing the area allocated to the high category and, to a lesser extent, to the very high and extreme categories.

Group 4 (the Sierras Grandes region) comprises mountainous environments with little human influence, characterised by the presence of high-mountain grasslands with extensive areas of fine fuel, seasonal accumulation of biomass, complex topography, and strong wind conditions. These conditions favour the rapid spread of fire and potentially extreme behaviour once a fire has started, regardless of the cause of ignition. Based on these conditions, one would expect an almost total concentration in the 'extreme' category and a very low representation of the remaining categories, as shown by both models.

Cluster 5 is characterised as a transitional environment located in the central-western part of the Province of Córdoba, where features of the plains and mountain ranges converge. It forms an intermediate zone between the Sierras Grandes and the eastern plains, encompassing mountainous areas, foothills, and gently undulating plains. From a territorial perspective, it exhibits a highly heterogeneous landscape, comprising a mosaic of natural vegetation, grasslands, and productive areas, alongside urban developments at various stages of consolidation. This configuration results in marked fragmentation and high variability in fuel continuity, which influences fire behaviour under local conditions and strongly affects its spread. The adjusted model tends to shift the probability towards lower danger categories, reducing the representation of the high and very high categories, while both models remain consistent under extreme conditions.

Finally, Cluster 7, located in the north-west of the province, corresponds to a predominantly arid environment, characterised by plains and low mountain ranges. It is distinguished by a continuous landscape of natural vegetation, dominated by xerophytic forests and scrubland, with little fragmentation and limited intensive human use, which may be related, among other factors, to the implementation of policies for the management and conservation of native forests. The abundance and continuity of woody fuels mean that fires, although less frequent, can reach high intensity and have significant impacts on largely natural ecosystems. For these reasons, the 'extreme' category is expected to be predominant, as indicated by both models. However, the adjusted model significantly reduces the probability of the 'high' and 'very high' categories; to some extent, the 'very high' category is effectively replaced by the 'high' category in the new model.

In this context, and to evaluate the spatial consistency and operational performance of the proposed approach, Figures 10, 11, and 12 present a direct comparison between the standard FWI classification and the percentile-based classification for three representative days.

Across all three cases, a consistent pattern emerges: the standard classification systematically overestimates the spatial extent of high and extreme fire danger categories. This results in a relatively homogeneous representation of fire danger, which fails to adequately capture the environmental heterogeneity of the region. In contrast, the percentile-based approach introduces a more spatially differentiated distribution, reducing the dominance of extreme categories while

increasing the representation of moderate and high danger levels.

In Figure 10, the standard FWI classification shows a widespread dominance of high and very high danger across large portions of the province, highlighting the limited ability of fixed thresholds to represent spatial variability. Conversely, the percentile-based classification reveals a more structured spatial pattern, in which danger levels are distributed according to the previously identified environmental regions. This redistribution is particularly evident in transitional areas, where zones originally classified as high and very high are reassigned to moderate category, in agreement with local environmental conditions.

Figure 11 reinforces this behaviour under different meteorological conditions. Although both approaches respond to variations in fire weather, the percentile-based classification maintains a more balanced distribution of danger levels. In particular, high and very high danger becomes more spatially constrained and concentrated in regions where environmental conditions support high fire potential.

In Figure 12, the differences between both approaches become even more evident in terms of spatial coherence. The standard classification produces large, continuous areas of high and extreme danger, potentially leading to an overgeneralised perception of fire risk. In contrast, the percentile-based classification introduces greater fragmentation and gradation in danger levels, which is more consistent with the influence of environmental characteristics. This effect is particularly relevant in heterogeneous landscapes, such as agricultural or peri-urban areas.

From an operational perspective, these results have important implications. The reduction in the spatial extent of extreme categories helps mitigate the problem of overestimation by avoiding the systematic triggering of high-level alerts. At the same time, the percentile-based approach preserves the identification of critical conditions by concentrating extreme danger in areas where the combination of meteorological and environmental factors justifies it. In addition, the resulting classification exhibits more gradual transitions between adjacent danger categories, providing a smoother and more realistic representation of spatial variability in fire risk.

Overall, Figures 10–12 provide strong support for the proposed methodology. The percentile-based classification reduces the bias associated with fixed thresholds and enhances the spatial realism of fire danger representation. This improvement is consistent across different days and meteorological scenarios, reinforcing the robustness and operational applicability of the approach in heterogeneous environments such as the Province of Córdoba.

6. Conclusions

The proposed approach addressed the critical limitations of the Canadian Fire Weather Index (FWI) when applied with standard thresholds in the Province of Córdoba, Argentina. The results demonstrate that a uniform threshold approach leads to a systematic overestimation of fire danger,

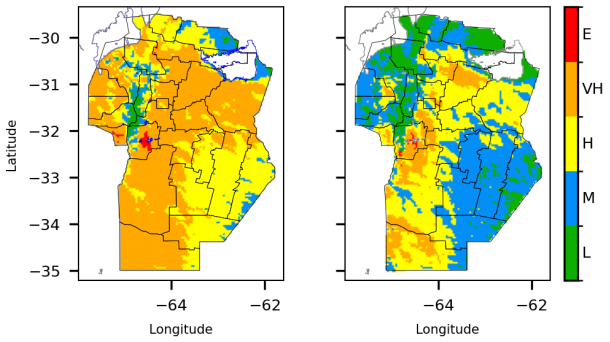


Figure 10: Comparison of fire danger classification based on the FWI. Left: standard classification using fixed thresholds. Right: percentile-based classification proposed in this study. The figure highlights differences in the spatial distribution of fire danger levels across Córdoba.

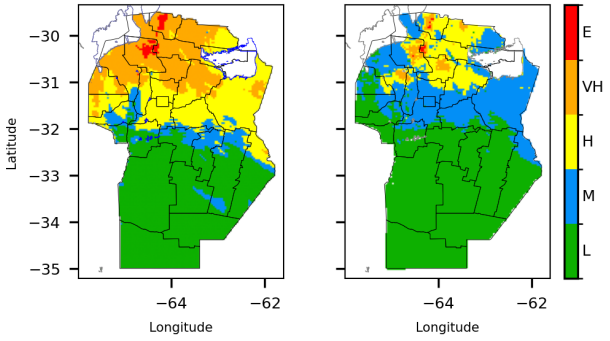


Figure 11: Same comparison as in Figure 10, for a different day. The percentile-based classification shows a redistribution of fire danger levels, reducing the spatial extent of extreme categories relative to the standard approach.

potentially causing "warning fatigue" among civil protection organisations and undermining operational reliability [10, 22]. To address these limitations, a spatially explicit calibration framework was developed that integrates environmental heterogeneity through k-means clustering and percentile-based threshold estimation.

The regionalisation process identified seven environmentally homogeneous regions, effectively capturing the complex gradients of climate, topography, vegetation, and anthropogenic influence across the province. Within these regions, the redefinition of FWI thresholds using local percentiles derived from WRF model outputs showed a marked improvement in fire danger representation. Specifically, the statistical evaluation provides robust evidence for the superiority of Accumulated BA over Hotspot Counts (HS) as a validation metric. In the critical 60th–95th percentile range, the BA-based calibration achieved a significantly lower RMSE (0.073 vs 0.113) and a higher R^2 value (0.460 vs 0.382) compared to filtered hotspots. These findings confirm that BA acts as a physical filter, effectively isolating the fire-weather signal from the “ignition noise” caused by anthropogenic activities in low-to-mid percentiles. The results also

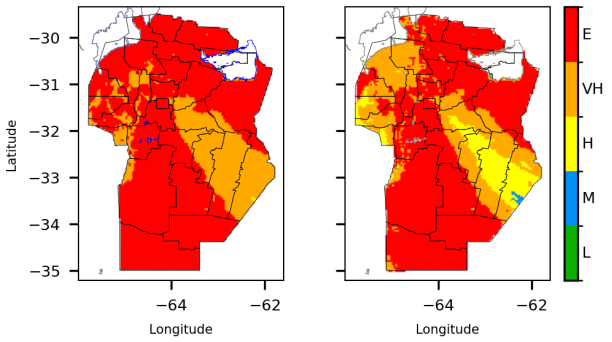


Figure 12: Comparison of FWI-based fire danger classification for another representative case. The percentile-based approach redistributes areas originally classified as high or extreme danger into lower categories, reflecting a more balanced representation of fire risk.

highlight a strongly non-linear relationship between FWI and fire impact, where 70.7% of the total BA is concentrated above the 80th percentile. This validates the 80th percentile as a robust indicator of critical fire conditions in Córdoba, focusing the early warning system on the high-impact events that truly challenge suppression resources.

Despite these advancements, it is acknowledged that the two-year calibration period (2023–2024) remains a temporal constraint. While the current results are operationally significant, the integration of longer historical time series as the OHMC system matures will be essential to achieve even greater statistical robustness and to better account for long-term interannual climatic variability.

Overall, this work demonstrates that incorporating local environmental variables into fire danger thresholds significantly enhances the predictive skill of the FWI. The proposed methodology is reproducible and designed for direct integration into operational systems based on numerical weather prediction models such as WRF. Finally, this framework is transferable to other topographically and ecologically complex regions, providing a scientifically rigorous basis for fire management and risk mitigation strategies globally.

CRediT authorship contribution statement

Leandro Maturano: Investigation, Data curation, Formal analysis. **Luz Fuentes:** original draft, Methodology, Software, Formal analysis. **Matías Ezequiel Suárez:** Review & Editing, Software, Formal analysis. **Sergio Masuelli:** Writing - Review & Editing, Translation. **Elena Lucía Sandalio:** Writing - Review & Editing, Translation. **Andrés Rodríguez:** Review. **Denis Poffo:** Visualization, Formal analysis, Writing - Original draft, Supervision, Conceptualization.

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Table 3
Statistical distribution of FWI values by percentiles (0–100) for the seven identified clusters in Córdoba. Values represent the thresholds for the June–November period (2023–2024). Note the significant threshold variation between the high-altitude regions (Cluster 4) and the arid plains (Cluster 7).

Percentile	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Cluster 6	Cluster 7
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5.0	1.336	3.170	2.056	0.080	1.775	0.755	8.058
10.0	5.283	7.228	6.126	0.966	6.027	3.425	15.976
15.0	8.800	11.176	10.086	3.424	9.809	5.321	20.069
20.0	12.229	13.953	13.395	5.562	13.239	7.116	23.269
25.0	15.259	16.124	16.162	7.706	16.168	9.050	26.007
30.0	17.929	18.077	18.597	9.651	18.676	11.059	28.575
35.0	20.331	19.935	20.962	11.518	21.146	13.033	31.191
40.0	22.584	21.934	23.248	13.318	23.600	14.949	33.943
45.0	24.755	24.036	25.360	15.159	26.106	16.957	36.710
50.0	26.978	26.146	27.546	17.213	28.718	18.938	39.652
55.0	29.361	28.357	30.014	19.393	31.443	21.090	42.771
60.0	32.024	30.665	32.733	21.811	34.304	23.371	45.966
65.0	35.087	33.416	35.790	24.362	37.319	25.930	49.421
70.0	38.656	36.632	39.266	27.233	40.636	28.526	53.311
75.0	43.053	40.007	43.296	30.424	44.487	31.493	58.130
80.0	48.620	43.842	48.115	34.285	48.899	35.141	63.811
85.0	55.268	48.452	53.798	39.250	54.354	40.067	71.079
90.0	63.394	54.523	60.560	46.698	61.829	47.057	80.613
95.0	75.282	63.941	70.494	59.606	74.272	58.041	96.161
100.0	214.225	142.794	197.540	225.044	267.345	129.369	225.618

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A. Supplementary Tables

A.1. Complete FWI percentile table